



VIDEO AUTOMATION

Kinetic Video & Audio Generation Pipeline

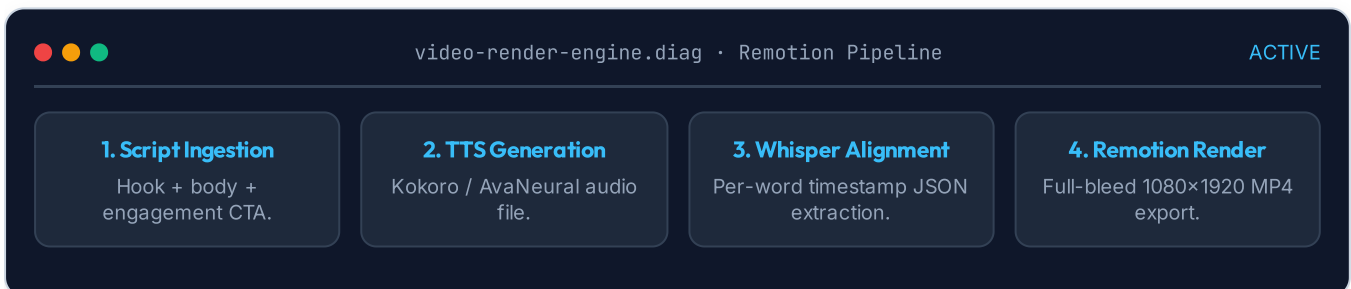
Automated 9:16 vertical short-form video compilation with Remotion React, TTS, and kinetic captions.

EXECUTIVE SUMMARY

Programmatic video engines enable instant generation of high-retention short-form videos for YouTube Shorts and TikTok. This technical blueprint covers Remotion React compositions, word-level audio synchronization, and audio mastering.

1 Automated 9:16 Video Composition Pipeline

Using Remotion, video timelines are defined as declarative React components. Dynamic media assets, background video loops, and word-level synchronized typography assemble automatically via CLI rendering.



2 Production Implementation & Schema

```
// Remotion Kinetic Captions Component (React)
export const KineticCaption = ({ words, currentTime }) => {
  const current = words.find(w => currentTime ≥ w.start && currentTime ≤ w.end);
  return (
    <div>
      {current?.text}
    </div>
  );
};
```



3 Audio Loudness Mastering (-14 LUFS)

Short-form platforms penalize inconsistent volume. All master audio passes through an FFmpeg loudnorm filter targeting exactly -14 LUFS with a -1.5 dB true peak to eliminate clipping.

remotion-composer · Render Log

CONNECTED

```
[REMOTION] Compiling vertical short (1080x1920, 30 FPS) ...
✓ [AUDIO] Voiceover generated · Integrated Loudness: -14.0 LUFS
✓ [CAPTIONS] 74 word timestamps aligned via Whisper
✓ [RENDER] 900 frames compiled in 11.4s · Exported to Desktop
[SAVED] C:\Users\futur\Desktop\Pinnacle_Short_01.mp4
```

4 Key Takeaways & Best Practices

- ✓ **Sub-3 Second Hook**

Place primary visual movement and high-contrast headlines within the first 90 frames to maximize viewer retention.
- ✓ **High-Contrast Captions**

Use bold uppercase typography with electric gold active-word highlights positioned inside vertical safe zones.
- ✓ **Background Music Mixing**

Set background audio tracks at volume=0.32 relative to voiceover to prevent frequency clash.
- ✓ **Zero Media Bleed**

Ensure video b-roll tracks are stripped of original background noise before final multi-track render.

IMPLEMENTATION SUPPORT

Need engineering assistance integrating these architectures into your business? Contact our team directly at futureai4all@gmail.com.